

Task Management System — End-to-End Requirement Analysis



Project Overview

The Task Management System is a role-based web portal that replaces manual, paper/WhatsApp-driven task tracking within an institution (college/department setup) with a digitized workflow. It connects three tiers – Administrator → Staff → Student – through a task delegation chain, automated deadline enforcement, file-based proof submission, and analytics dashboards.

 **Core idea:** Admin assigns tasks to Staff → Staff breaks them down and assigns to Students → Students submit proof of completion → Everyone gets automated email nudges before deadlines slip.

Objectives

- Digitize manual task assignment and tracking processes
- Enforce role-based access control (RBAC) across three user tiers
- Automate deadline monitoring with pre-deadline warning emails
- Provide a centralized proof-of-completion repository (certificates, assignment files, event proof)
- Give Admin and Staff visibility into team/individual workload through dashboards
- Enable a structured query/escalation channel (Staff → Admin)
- Generate analytical reports on task completion, delays, and performance

Stakeholders & Roles

Role	Reports To	Manages	Access Level
Administrator	—	Staff (and visibility into Students)	Full system access
Staff	Administrator	Students	Mid-level – own team only
Student	Staff	—	Restricted – own tasks only

Functional Requirements by Role

4.1 Administrator

Req.	Requirement
A1	Create, update, and deactivate Staff and Student accounts
A2	View full profile and activity history of every Staff and Student
A3	Create tasks and assign them only to Staff (not directly to Students)
A4	Set task title, description, category, priority, and deadline
A5	Receive status intimation when a Staff member marks a task complete
A6	Receive and resolve queries/escalations raised by Staff
A7	Receive automated warning email when a Staff member is approaching a missed deadline (before the stipulated time)
A8	View a dashboard showing all Staff under them, with task counts (pending/in-progress/completed/overdue) per Staff
A9	Generate and export analytical reports (completion rate, average delay, staff-wise performance)
A10	View audit trail of key actions (task creation, reassignment, approvals)

Functional Requirements by Role

4.2 Staff

Req.	Requirement
S1	View tasks assigned by Admin
S2	Break down / translate an Admin task into one or more student-facing tasks (e.g., "complete this course," "submit this assignment")
S3	Assign tasks to one or more Students
S4	Mark a task complete and intimate Admin
S5	Raise a query to Admin if clarification is needed on a task
S6	Receive automated warning email if their own task is nearing its deadline without completion
S7	Receive completion intimation when a Student finishes an assigned task
S8	Review submitted proof (course certificate, assignment file, event proof) and leave a review/remark
S9	Approve or reject a student submission; rejected submissions should support resubmission
S10	View a dashboard of all Students under them, with individual task status
S11	System automatically emails a warning to the Student before their stipulated deadline if the task is still pending

4.3 Student

Req.	Requirement
U1	View tasks assigned by their Staff supervisor
U2	Upload task output/proof as a file (assignment document, course completion certificate, event attendance proof, etc.)
U3	View status of each submission (Pending / Under Review / Approved / Rejected)
U4	View Staff's review/remarks on a submission
U5	Resubmit a task if rejected
U6	Receive automated warning email before the task deadline if not yet submitted
U7	View personal task history and deadlines in one place (No admin/staff-level access – view and upload only)

System Modules

5.1 Authentication & Authorization Module

- Secure login (email/password, institutional SSO optional)
- Role-based access control (Admin / Staff / Student)
- Password reset, session management, account lock on suspicious activity

5.2 User Management Module (Admin only)

- Create/edit/deactivate Staff and Student accounts
- Assign Students to a Staff supervisor; assign Staff under Admin
- Bulk import (CSV) for onboarding a batch of students/staff

5.3 Task Management Module

- Task creation with title, description, category, priority, deadline, attachments
- Task categories: Course Completion, Subject Assignment, Monthly Meeting, Event Attendance (with proof), and extensible custom categories
- Task delegation chain: Admin → Staff → Student
- Task status lifecycle: Created → Assigned → In Progress → Submitted → Under Review → Approved/Rejected → Closed
- Reassignment and deadline-extension request support

5.4 Submission & File Management Module

- Student/Staff file upload (PDF, image, docx) as completion proof
- Version history if resubmitted after rejection
- Secure file storage with access restricted to relevant Admin/Staff/Student only

5.5 Review & Approval Module

- Staff reviews Student submissions, adds remarks, approves/rejects
- Admin reviews Staff completion intimation
- Review comments visible to the submitter

System Modules (Continued)

5.6 Notification & Automated Escalation Module

This is the automation core of the system:

- Scheduled job checks all open tasks daily/hourly against their deadlines
- If a task is still open at a configurable threshold before the deadline (e.g., 24–48 hrs prior), the system auto-emails a warning to the responsible person: Staff → warning email if their Admin-assigned task is at risk; Student → warning email if their Staff-assigned task is at risk
- Completion intimations flow upward automatically (Student completes → Staff notified; Staff completes → Admin notified)
- Query/escalation notifications routed to Admin in real time
- In-app notification bell in addition to email (recommended)

5.7 Query / Escalation Module

- Staff can raise a query against a specific task, addressed to Admin
- Ticket-style status: Open → Responded → Resolved
- Admin response visible to Staff in-thread

5.8 Dashboard & Analytics Module

- Admin dashboard: total Staff, tasks per Staff, overdue task count, org-wide completion %
- Staff dashboard: total Students, tasks per Student, pending reviews, overdue count
- Student dashboard: personal task list, upcoming deadlines, submission history
- Visual charts: status breakdown, on-time vs delayed completion trend, workload distribution

5.9 Reporting Module

- Exportable reports (PDF/Excel): completion rate, average turnaround time, delay frequency, staff/student performance
- Filterable by date range, task category, role, or individual

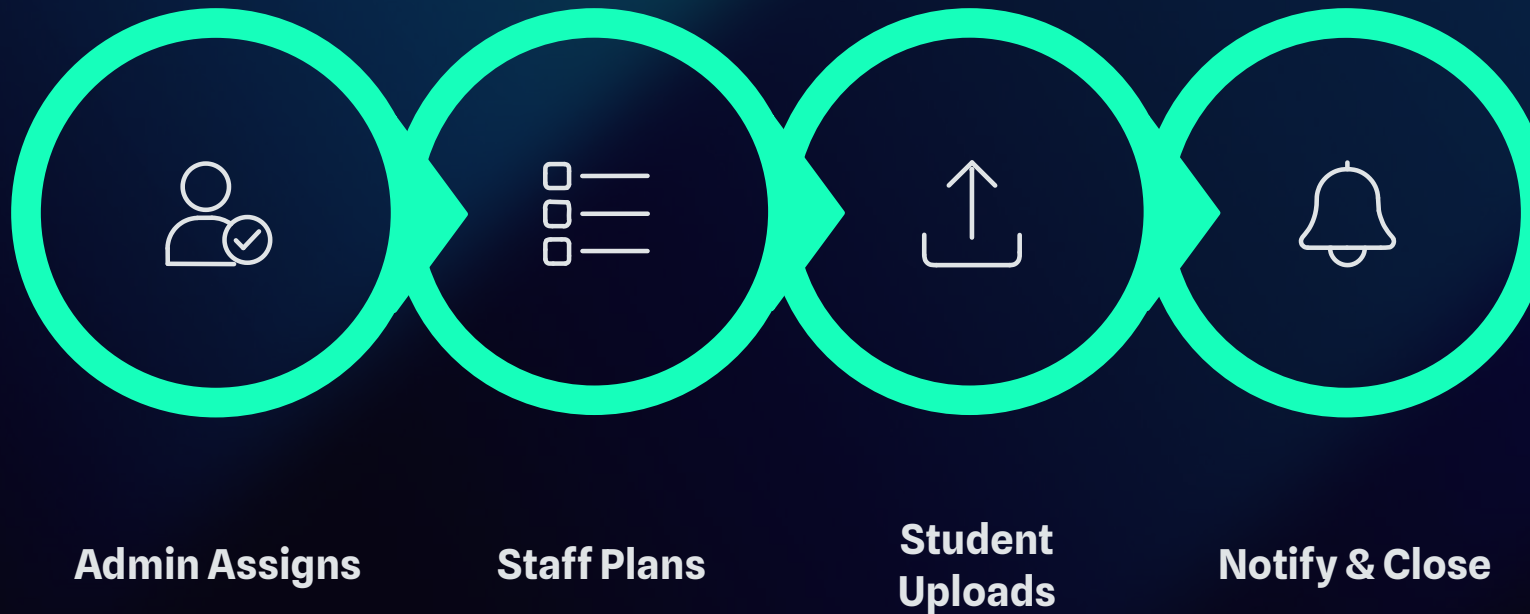
5.10 Audit & Logging Module

- Immutable log of task creation, assignment, status changes, approvals/rejections, and login activity
- Useful for institutional accountability and dispute resolution

Non-Functional Requirements

Category	Requirement
Security	RBAC enforcement at API level (not just UI hiding); encrypted passwords (bcrypt/argon2); HTTPS everywhere; file upload validation to block malicious files
Integration	Secure integration with institutional systems (e.g., LDAP/Active Directory or Google Workspace SSO for login; SMTP relay for institutional email)
Reliability	Scheduled notification jobs must be fault-tolerant (retry on failure, no duplicate/missed alerts)
Scalability	Should handle growth from a single department to institution-wide rollout without redesign
Usability	Responsive UI (desktop + mobile), minimal clicks for Student file upload
Auditability	All status-changing actions logged with timestamp and actor
Data Privacy	Certificates/personal files access-restricted to relevant chain (Student ↔ Staff ↔ Admin only)
Availability	Target uptime suitable for academic-cycle-critical usage (deadlines, submissions)

Core Workflow (Task Lifecycle)



Admin creates a task and assigns it to a Staff member with a deadline. Staff receives it, converts it into one or more sub-tasks, and assigns them to relevant Students. System schedules a pre-deadline reminder for both the Staff-level and Student-level deadlines. Student completes the work and uploads proof (file/certificate). Staff reviews the submission → Approves (task closed, Staff notified upward) or Rejects (Student notified, resubmission allowed). Once all sub-tasks under a Staff member are approved, Staff marks the parent task complete → Admin is intimated. If at any point the Staff or Student misses the approaching deadline, the system auto-sends a warning email before the stipulated time – not after. If Staff has a doubt about a task at any stage, they raise a query to Admin, which appears in the Admin's query queue until resolved. Admin dashboard aggregates all of this into live counts: Staff under them, tasks per Staff, overdue items, resolved queries.

High-Level Data Model

Entity	Key Fields
Users	id, name, email, password_hash, role (admin/staff/student), supervisor_id, department, status
Tasks	id, title, description, category, priority, deadline, created_by (admin), assigned_to (staff), status, parent_task_id (for staff→student breakdown)
Submissions	id, task_id, submitted_by (student), file_url, submitted_at, status, review_remarks, reviewed_by
Notifications	id, user_id, type (warning/completion/query), message, sent_at, read_status
Queries	id, raised_by (staff), raised_to (admin), task_id, subject, message, status, resolved_at
AuditLogs	id, user_id, action, entity_type, entity_id, timestamp

Relationships

Admin → Staff
One Admin → many Staff (1:N)

Staff → Students
One Staff → many Students (1:N)

Task Cascade
One Task can cascade into many child Tasks (Staff-created, linked via parent_task_id)

Task → Submissions
One Task → many Submissions (in case of resubmission)

Suggested Technology Stack

Layer	Recommendation
Frontend	Next.js (React) + Tailwind CSS
Backend	Next.js API routes or a separate Node.js/Express service
Database	PostgreSQL or MySQL (relational fits the strict role/task hierarchy well)
Auth	NextAuth.js or JWT-based auth, with optional LDAP/SSO connector
File Storage	AWS S3 / Cloudinary / Firebase Storage (for certificates & submission files)
Scheduled Jobs	node-cron, or a serverless cron (Vercel Cron / AWS EventBridge) for deadline checks
Email	Nodemailer via institutional SMTP or a transactional email service (SendGrid/SES)
Real-time (optional)	Socket.io for in-app notification bell
Deployment	Vercel / Render / institutional server

Assumptions, Constraints & Future Enhancements

Assumptions & Constraints

Each Student has exactly one Staff supervisor at a time; each Staff reports to one Admin (simple hierarchy – can be extended to multi-admin later).

Only Admin can create top-level tasks; Staff cannot originate tasks independently of an Admin-issued task.

Email is the primary notification channel; institutional mail server access is available for sending automated alerts.

"Stipulated time" warning threshold (e.g., 24 hrs before deadline) is configurable, not hardcoded.

File uploads are limited to common formats (PDF/JPG/PNG/DOCX) with a reasonable size cap.

Future Enhancements (Out of Scope for v1)

Mobile App

Mobile app companion

SMS / WhatsApp

SMS/WhatsApp alerts alongside email

Multi-Admin Hierarchy

Multi-admin / multi-department hierarchy

AI-Assisted Review

AI-assisted review suggestions on submitted certificates

Calendar Sync

Calendar sync (Google Calendar) for deadlines